

SDP 302

Problem Set: Monte Carlo Method & OLS

Professor: Michael Wegener | Term: Fall 2026 |

Instructions & Information

The **Monte Carlo Method** is an elegant and powerful computational technique that uses repeated sampling to solve complex problems with uncertain variables the solutions of which are difficult or impossible to calculate deterministically. Instead of using rigid analytical methods to find a single, exact deterministic solution to the problem, the Monte Carlo method aims to estimate the outcome of uncertain events by running a plethora of repeated random samplings to provide a range of possible outcomes along with the probability of each occurring. The method goes back to Metropolis and Ulam 1949. Named after the famous casino in Monte Carlo due to its heavy reliance on chance and probability, Monte-Carlo techniques are widely considered a cornerstone in Bayesian statistics and modern data science. Due to their simplicity and versatility they have also become a key method in quantitative research in many scientific fields. The book by Fishman 1996 offers an excellent introduction to Monte Carlo methods, providing both theoretical foundations and practical applications.

In this assignment we will use the Monte Carlo method to demonstrate the following key theoretical properties of the Ordinary Least Squares (OLS) estimator under standard Gauss-Markov assumptions:

1. Unbiasedness: $\mathbb{E}[\hat{\beta}_i] = \beta_i$, i.e. on average, the estimated coefficients equal the true population parameters.
2. Consistency and decreasing variance: As sample size N increases, the distribution of $\hat{\beta}$ collapses around the true value.
3. Sampling distribution and normality: According to the central limit theorem, the estimators $\hat{\beta}_i$ follow a normal distribution in large samples even when error terms are non-normal.

As a starting and later reference point we assume to know the true population model which we define as:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i$$

where true parameters are $\beta_0 = 1$, $\beta_1 = -3$, $\beta_2 = 2$, and errors are i.i.d. and either $\epsilon_i \sim \mathcal{N}(0, 6)$ or $\epsilon_i \sim \mathcal{U}(0, 6)$. This true regression model serves as our data generating process (DGP), which we can use to generate artificial samples of different size N in M independent simulation runs. For each sample size we can calculate the OLS estimators and compare them against the benchmark of the known true population parameters.

References

- Fishman, George S. (1996). *Monte Carlo: Concepts, Algorithms, and Applications*. Springer Series in Operations Research. Springer.
- Metropolis, Nicholas and S. Ulam (1949). "The Monte Carlo Method". In: *Journal of the American Statistical Association* 44.247, pp. 335–341.

[EXR] Exercise 1 (Exploring the BLUE Properties of OLS)

In the *GitHub* repository for this course you will find the files *MonteCarlo_BLUE.csv* and *MonteCarlo_BLUE.parquet*. This file contains the OLS estimators that were generated in 10,000 simulation runs for each of the following sample sizes: $N = 15$, $N = 50$, $N = 200$, $N = 1,000$. Import the file as a pandas DataFrame and answer the questions below. Note that the last column of the DataFrame contains a binary categorical variable that can be used to filter the data for the distribution of the error terms (1 for normal distribution, 0 for uniform distribution).

- a) Explain why OLS linear regression remains valid for estimating β_0, β_1 , and β_2 despite the non-linear term X^2 in the population model.
- b) Filter the DataFrame for those samples in which the error term followed a normal distribution. For each sample size, calculate the mean of each coefficient's 10,000 estimates and compare this average with the true value. Also plot a separate histogram for each sample size illustrating the distribution of each estimator and mark the true value on each plot. What do you notice about the centre, the spread and shape of these distributions as the sample size increases? Relate your findings and conclusion to the three properties listed above.
- c) Filter the DataFrame for those samples in which the error terms were i.i.d. but followed the uniform distribution. Repeat the above instructions for these samples. Do your findings and conclusions change? If yes, what do you think affects these findings?